

Homework Assignment 3 - Coding Part Write-up

Networks and Markets

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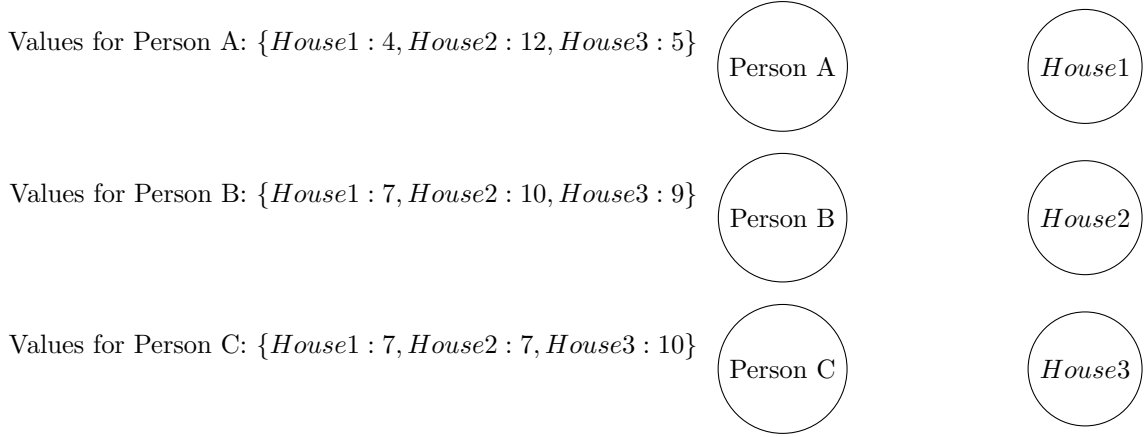
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Part 4: Implementing Matching Market Pricing

1 Question 7

(b) Consider the matching market example in Lecture 5 Page 7:



Formally, the matching market context is $\Gamma = (\{A, B, C\}, \{1, 2, 3\}, v)$, where v is the valuation function defined as follows:

$$\begin{aligned}v_A(1) &= 4, v_A(2) = 12, v_A(3) = 5 \\v_B(1) &= 7, v_B(2) = 10, v_B(3) = 9 \\v_C(1) &= 7, v_C(2) = 7, v_C(3) = 10\end{aligned}$$

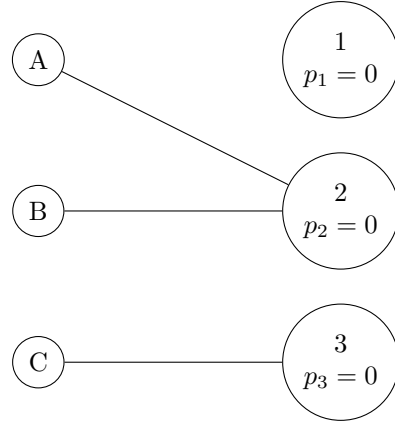
We turn to run the algorithm of Theorem 8.8 to find a market equilibrium (p, M) to find the maximum social value, in order to validate our implementation's output. We begin by initializing the prices vector $\vec{p} \equiv 0$ to be the zero vector. We then proceed to run the algorithm, updating the prices vector until there is a perfect matching M in the induced preferred choice graph for $(\Gamma, \vec{p})$:

1. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:

Values for A : $\{1 : 4, 2 : 12, 3 : 5\}$
 Utility for A : $\{1 : 4, 2 : \mathbf{12}, 3 : 5\}$

Values for B : $\{1 : 7, 2 : 10, 3 : 9\}$
 Utility for B : $\{1 : 7, 2 : \mathbf{10}, 3 : 9\}$

Values for C : $\{1 : 7, 2 : 7, 3 : 10\}$
 Utility for C : $\{1 : 7, 2 : 7, 3 : \mathbf{10}\}$



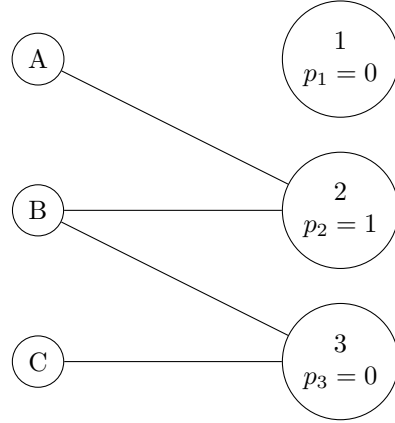
There obviously isn't a perfect matching as $S = \{A, B\}$ is a constricted set with $|N(S)| = |\{2\}| = 1 < 2 = |S|$ (which, by a theorem we've seen in class, implies that there isn't a perfect matching). Thus, we raise the prices for all items in $N(S)$ by 1, and update the prices vector $\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 1, c : 0)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

2. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:

Values for A : $\{1 : 4, 2 : 12, 3 : 5\}$
 Utility for A : $\{1 : 4, 2 : \mathbf{11}, 3 : 5\}$

Values for B : $\{1 : 7, 2 : 10, 3 : 9\}$
 Utility for B : $\{1 : 7, 2 : \mathbf{9}, 3 : \mathbf{9}\}$

Values for C : $\{1 : 7, 2 : 7, 3 : 10\}$
 Utility for C : $\{1 : 7, 2 : 6, 3 : \mathbf{10}\}$



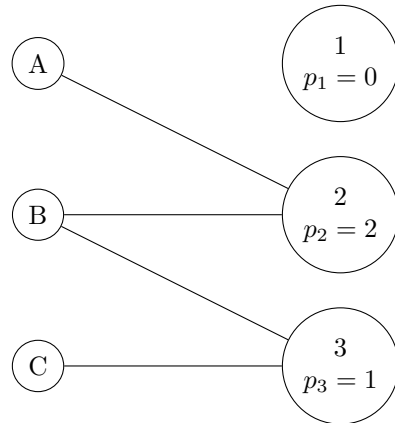
There obviously isn't a perfect matching as $S = \{A, B, C\}$ is a constricted set with $|N(S)| = |\{2, 3\}| = 2 < 3 = |S|$ (which, by a theorem we've seen in class, implies that there isn't a perfect matching). Thus, we raise the prices for all items in $N(S)$ by 1, and update the prices vector $\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 2, c : 1)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

3. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:

Values for A : $\{1 : 4, 2 : 12, 3 : 5\}$
 Utility for A : $\{1 : 4, 2 : \mathbf{10}, 3 : 4\}$

Values for B : $\{1 : 7, 2 : 10, 3 : 9\}$
 Utility for B : $\{1 : 7, 2 : \mathbf{8}, 3 : \mathbf{8}\}$

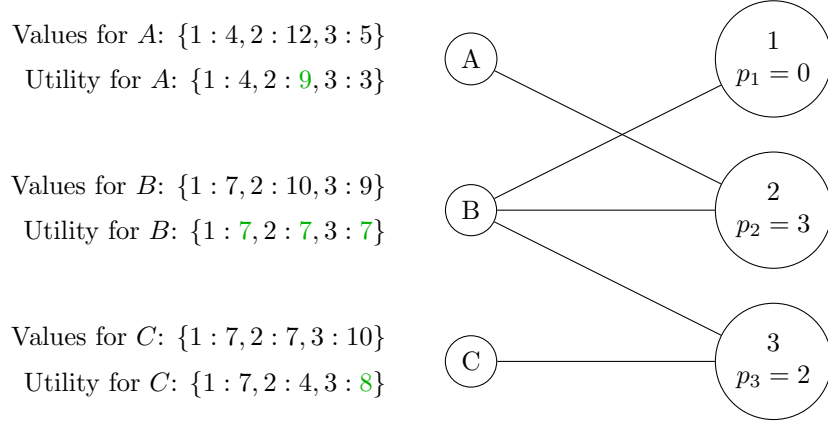
Values for C : $\{1 : 7, 2 : 7, 3 : 10\}$
 Utility for C : $\{1 : 7, 2 : 5, 3 : \mathbf{9}\}$



Similar to the previous iteration, we raise the prices for $\{2, 3\}$, and update the prices vector

$\vec{p}$ accordingly. The updated prices vector is $\vec{p} = (a : 0, b : 3, c : 2)$. Not all prices are greater than zero, so we don't perform a shift operation, and we proceed to the next iteration.

4. Observing the following *induced preferred-choice graph* from $(\Gamma, \vec{p})$:



And there is a perfect matching in the induced preferred choice graph, which is $M = \{\{A, 2\}, \{B, 1\}, \{C, 3\}\}$. Thus, the market equilibrium is $(\vec{p}, M) = ((1 : 0, 2 : 3, 3 : 2), \{\{A, 2\}, \{B, 1\}, \{C, 3\}\})$, and we are done.

We found the market equilibrium to be $(\vec{p}, M) = ((1 : 0, 2 : 3, 3 : 2), \{\{A, 2\}, \{B, 1\}, \{C, 3\}\})$. The maximum social value is therefore $v(A, 2) + v(B, 1) + v(C, 3) = 12 + 7 + 10 = 29$.

Our algorithm found exactly this market equilibrium.

2 Question 8

- (a) In this part we analyze how the prices output by the VCG mechanism compare with the ones output by the algorithm of Theorem 8.8 (finding a market equilibrium (p, M)). The following are the examples we analyze and their corresponding results for each mechanism:

1. Example 1:

Values for b_1 : $\{i_1 : 3, i_2 : 0, i_3 : 15\}$

b_1

i_1

Values for b_2 : $\{i_1 : 11, i_2 : 12, i_3 : 8\}$

b_2

i_2

Values for b_3 : $\{i_1 : 14, i_2 : 13, i_3 : 18\}$

b_3

i_3

Values for b_4 : $\{i_1 : 3, i_2 : 18, i_3 : 5\}$

b_4

Values for b_5 : $\{i_1 : 15, i_2 : 9, i_3 : 1\}$

b_5

Values for b_6 : $\{i_1 : 14, i_2 : 17, i_3 : 9\}$

b_6

Values for b_7 : $\{i_1 : 14, i_2 : 17, i_3 : 13\}$

b_7

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

2. Example 2:

Values for b_1 : $\{i_1 : 12, i_2 : 14, i_3 : 16, i_4 : 8, i_5 : 6, i_6 : 17\}$

b_1

i_1

Values for b_2 : $\{i_1 : 11, i_2 : 7, i_3 : 9, i_4 : 19, i_5 : 1, i_6 : 11\}$

b_2

i_2

Values for b_3 : $\{i_1 : 18, i_2 : 13, i_3 : 17, i_4 : 17, i_5 : 2, i_6 : 16\}$

b_3

i_3

Values for b_4 : $\{i_1 : 15, i_2 : 0, i_3 : 4, i_4 : 1, i_5 : 15, i_6 : 15\}$

b_4

i_4

Values for b_5 : $\{i_1 : 7, i_2 : 8, i_3 : 5, i_4 : 12, i_5 : 18, i_6 : 13\}$

b_5

i_5

Values for b_6 : $\{i_1 : 7, i_2 : 19, i_3 : 8, i_4 : 12, i_5 : 4, i_6 : 1\}$

b_6

i_6

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

3. Example 3:

Values for b_1 : $\{i_1 : 8, i_2 : 11, i_3 : 0, i_4 : 3, i_5 : 6, i_6 : 7\}$

b_1

i_1

Values for b_2 : $\{i_1 : 19, i_2 : 14, i_3 : 15, i_4 : 14, i_5 : 14, i_6 : 16\}$

b_2

i_2

Values for b_3 : $\{i_1 : 17, i_2 : 19, i_3 : 19, i_4 : 13, i_5 : 8, i_6 : 17\}$

b_3

i_3

Values for b_4 : $\{i_1 : 2, i_2 : 15, i_3 : 1, i_4 : 18, i_5 : 11, i_6 : 10\}$

b_4

i_4

Values for b_5 : $\{i_1 : 8, i_2 : 9, i_3 : 7, i_4 : 15, i_5 : 6, i_6 : 10\}$

b_5

i_5

Values for b_6 : $\{i_1 : 12, i_2 : 15, i_3 : 15, i_4 : 8, i_5 : 2, i_6 : 1\}$

b_6

i_6

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

4. Example 4:

Values for b_1 : $\{i_1 : 5, i_2 : 3, i_3 : 0, i_4 : 7, i_5 : 10, i_6 : 5, i_7 : 17, i_8 : 6, i_9 : 18, i_{10} : 8\}$

b_1

i_1

Values for b_2 : $\{i_1 : 5, i_2 : 4, i_3 : 6, i_4 : 9, i_5 : 15, i_6 : 9, i_7 : 17, i_8 : 2, i_9 : 10, i_{10} : 14\}$

b_2

i_2

Values for b_3 : $\{i_1 : 10, i_2 : 11, i_3 : 10, i_4 : 6, i_5 : 4, i_6 : 10, i_7 : 16, i_8 : 11, i_9 : 10, i_{10} : 6\}$

b_3

i_3

Values for b_4 : $\{i_1 : 2, i_2 : 19, i_3 : 4, i_4 : 12, i_5 : 5, i_6 : 8, i_7 : 12, i_8 : 0, i_9 : 11, i_{10} : 11\}$

b_4

i_4

Values for b_5 : $\{i_1 : 18, i_2 : 7, i_3 : 15, i_4 : 11, i_5 : 7, i_6 : 4, i_7 : 2, i_8 : 9, i_9 : 9, i_{10} : 8\}$

b_5

i_5

Values for b_6 : $\{i_1 : 5, i_2 : 2, i_3 : 2, i_4 : 5, i_5 : 1, i_6 : 12, i_7 : 13, i_8 : 18, i_9 : 8, i_{10} : 1\}$

b_6

i_6

i_7

i_8

i_9

i_{10}

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

5. Example 5:

Values for b_1 : $\{i_1 : 15, i_2 : 3, i_3 : 1, i_4 : 5\}$

b_1

i_1

Values for b_2 : $\{i_1 : 11, i_2 : 11, i_3 : 16, i_4 : 5\}$

b_2

i_2

Values for b_3 : $\{i_1 : 9, i_2 : 15, i_3 : 13, i_4 : 17\}$

b_3

i_3

Values for b_4 : $\{i_1 : 15, i_2 : 11, i_3 : 10, i_4 : 16\}$

b_4

i_4

Values for b_5 : $\{i_1 : 19, i_2 : 0, i_3 : 12, i_4 : 7\}$

b_5

Values for b_6 : $\{i_1 : 17, i_2 : 16, i_3 : 13, i_4 : 9\}$

b_6

And we observe that the prices output by the VCG mechanism and the algorithm of Theorem 8.8 are the same (the matching is also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism)

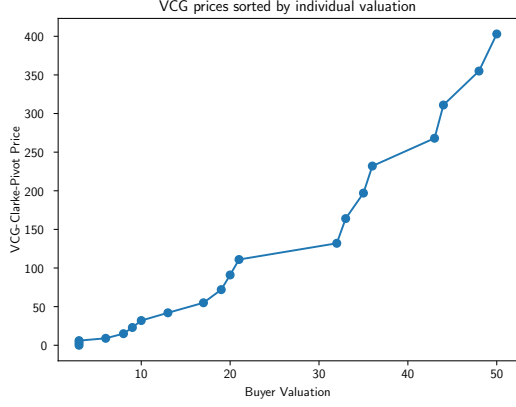
That is, in all examples we analyzed, the prices output by the VCG mechanism and the algorithm of Theorem 8.8 were the same, and the matching was also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism. We analyzed far more examples besides the ones presented here, and the results were consistent across all of them—the prices output by the VCG mechanism and the algorithm of Theorem 8.8 were the same (and the matching was also the same because we used the same algorithm to compute the socially optimal state as part of the VCG mechanism).

3 Bonus Question 2

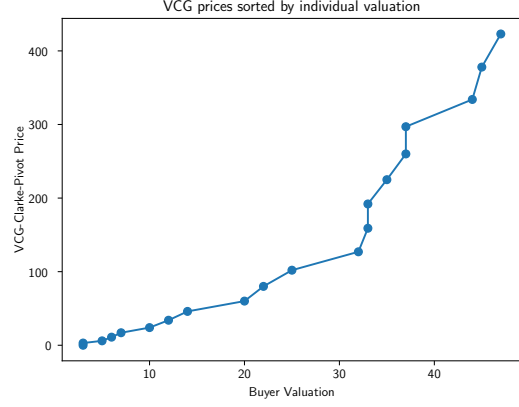
- (a) We structure a *markets-for-bundles* context of identical goods as a simple matching market context, where each bundle B_j 's value for bidder b_i is the product of the value of b_i for the good and the amount of goods in the bundle, c_i . That is, $v_i(B_j) = c_j \cdot t_i$, where t_i is the value of b_i for the singular good, and c_j is the amount of goods in bundle B_j . We then run the VCG algorithm we implemented in the previous part on a few randomized examples of such a markets-for-bundles context, where there are $n = m = 20$ bundles and bidders, and the values t_i are randomized between 1 and 50, and where $c_j = j$ ($j \in \{1, 2, \dots, 20\}$).

Figure 1 summarizes the results of the VCG algorithm on 4 such randomized examples of markets-for-bundles contexts (the results remained the same for other examples we ran). The x -axis represents the individual valuation t_i of the bidders for the singular good, and the y -axis represents the VCG price for the bidder—commonly referred to as the *externalities* of the bidder on the market.

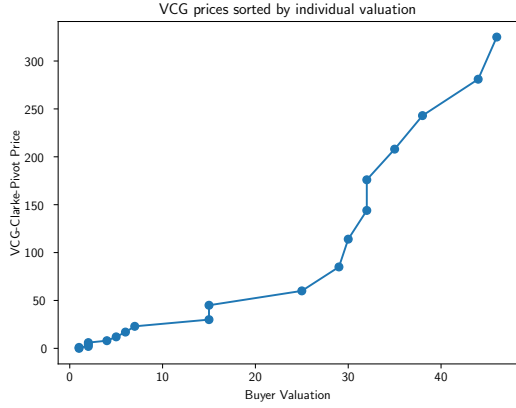
We observe a clear trend, where the VCG prices—i.e. *externalities* of bidders, are increasing with the valuation of the bidders for the singular good (post tie-breaks of valuations), which is expected given the structure of the markets-for-bundles context.



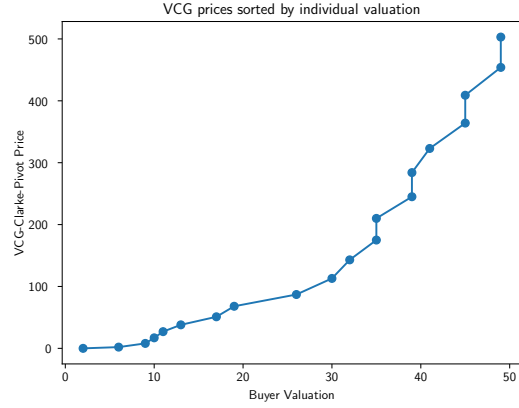
(a) Example 1



(b) Example 2



(c) Example 3



(d) Example 4

Figure 1: VCG prices for bidders in markets-for-bundles contexts

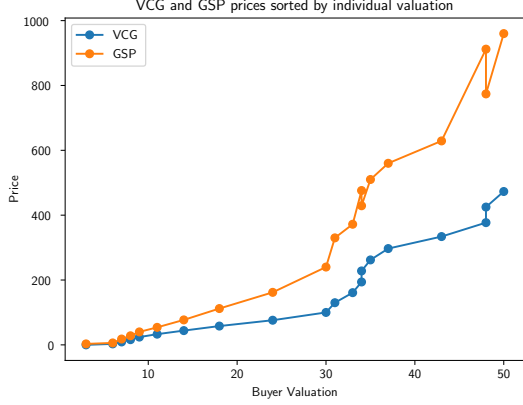
As we've seen in class, the socially optimal assignment in a *markets-for-bundles* of identical goods, is that the larger bundles are assigned to the higher valuation of a singular good. Thus, as the valuation of the singular good increases, the bidder b_i (w.l.o.g. the bidders are sorted by decreasing valuation of the singular goods) is assigned a larger bundle B_i (w.l.o.g. the bundles are sorted by decreasing sizes c_i). Without said bidder b_i , the bundles assignment shifts for the lower-bidding bidders. That is, $\forall j > i$, b_j is changed to be assigned B_{j-1} . This sets the externality to be $\sum_{j=i+1}^n (c_{j-1} - c_j) \cdot t_j$ for bidder b_i . In our case, this is $\sum_{j=i+1}^n t_j$. In either case, it is easy to see that as the valuation of the singular good increases (i.e., i grows), the externality of the bidder (i.e., of b_i) should increase for similarly distributed bundle sizes and similarly distributed valuations of the singular good (as in our randomization).

- (b) We implemented a GSP pricing mechanism in a *matching-market* context. The following are comparisons of the prices output by the GSP mechanism and the VCG mechanism over different contexts.

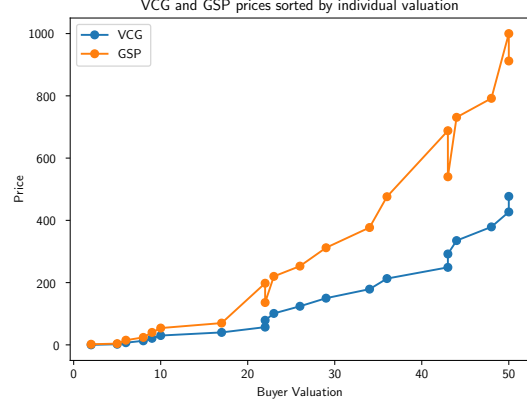
Randomized Matching-Market Contexts: We use the randomization scheme described in the previous part to generate randomized matching-market contexts, where there are $n = m = 20$ bidders and goods, and the values t_i are randomized between 1 and 50. We then run the GSP pricing mechanism we implemented on a few such randomized examples of matching-market contexts, and compare the prices output by the GSP mechanism to the VCG prices.

Figure 2 summarizes the results of the GSP pricing mechanism on 4 such randomized examples of matching-market contexts (the results remained the same for other examples we ran), compared to the VCG prices. The x -axis represents the individual valuation t_i of the bidders for the singular

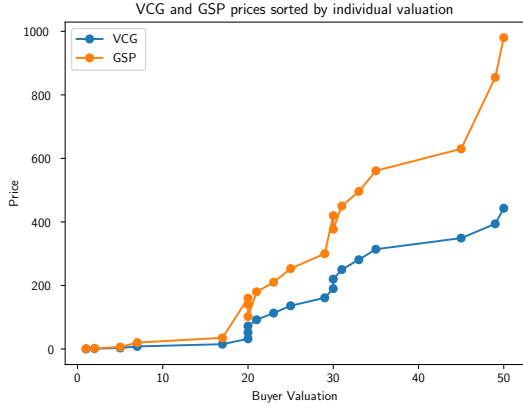
good, and the y -axis represents the GSP price for the bidder in orange and the VCG price for the bidder in blue.



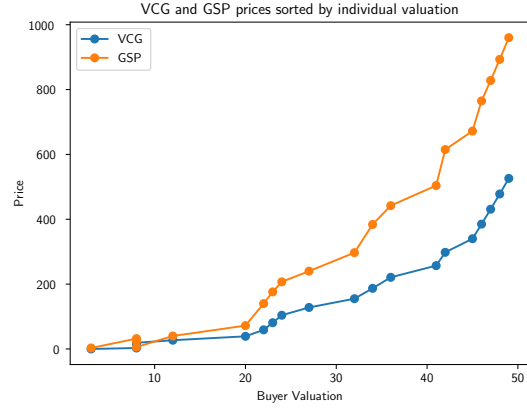
(a) Example 1



(b) Example 2



(c) Example 3



(d) Example 4

Figure 2: GSP and VCG prices for bidders in matching-market contexts

Unsurprisingly, the GSP prices are also increasing with the valuation of the bidders for the singular good (up to tie-breaks), as for every bidder b_i , the price/item is set to be the valuation of the next bidder b_{i+1} (or b_i if $i = n$), and the larger bundles are assigned to the higher-valuing bidders (as in the VCG mechanism). Thus, the GSP prices are increasing with the valuation of the bidders for the singular good, as the VCG prices are.

Contexts in which VCG and GSP prices are similar FILL IN

Contexts in which VCG and GSP prices are different FILL IN

(c)

(d)

Part 5: Exchange Networks for Uber

We will construct a simplified market scenario for a ridesharing app like Uber. Our world will consist of an $\ell \times \ell$ grid, and there will be two types of participants, riders and drivers.

- A rider R is specified by a current location $(x_0, y_0) \in [\ell] \times [\ell]$, a desired destination $(x_1, y_1) \in [\ell] \times [\ell]$, and a value for reaching that destination.

- A driver D is specified by a current location $(x_0, y_0) \in [\ell] \times [\ell]$.

We define the cost of a matching between a rider R and a driver D , $c(R, D)$, to be the distance from the driver to the rider and then to the destination (measured via Manhattan distance, i.e., L^1 distance).

1 Question 9

We are tasked with encoding the above as an exchange network. Namely, defining a graph $G = (V, E)$ where the vertices are the riders and rivers and there is en edge between every rider and driver.

Denoting the value for rider R reaching his destination as v_R , and reminded that the value of an edge in an exchange network must be a natural number, we define the values associated with each edge in the graph in the following way:

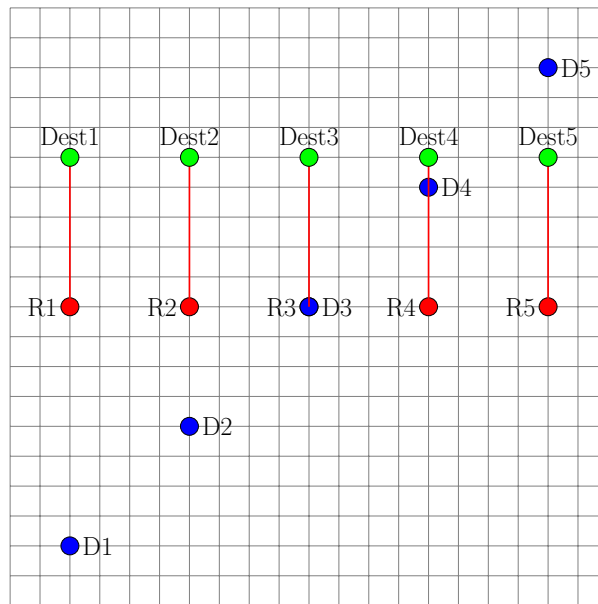
$$v(\overrightarrow{RD}) = \begin{cases} v_R - c(R, D) & \text{if } v_R \geq c(R, D) \\ 0 & \text{otherwise} \end{cases}$$

This value function ensures economic rationality by capping the cost of a match by the rider's valuation of reaching their destination—done by setting the value of the edge to zero if the rider's valuation is less than the cost of the match. This ensures that given any stable outcome, we can extrapolate a stable outcome where non-beneficial matches (i.e., where the value is realistically negative) are removed—where stability is maintained as the allocation of such rider-driver pairs would be 0 pre-removal (as the edge's value between them is 0) and 0 post-removal (as unmatched nodes have a value of 0).

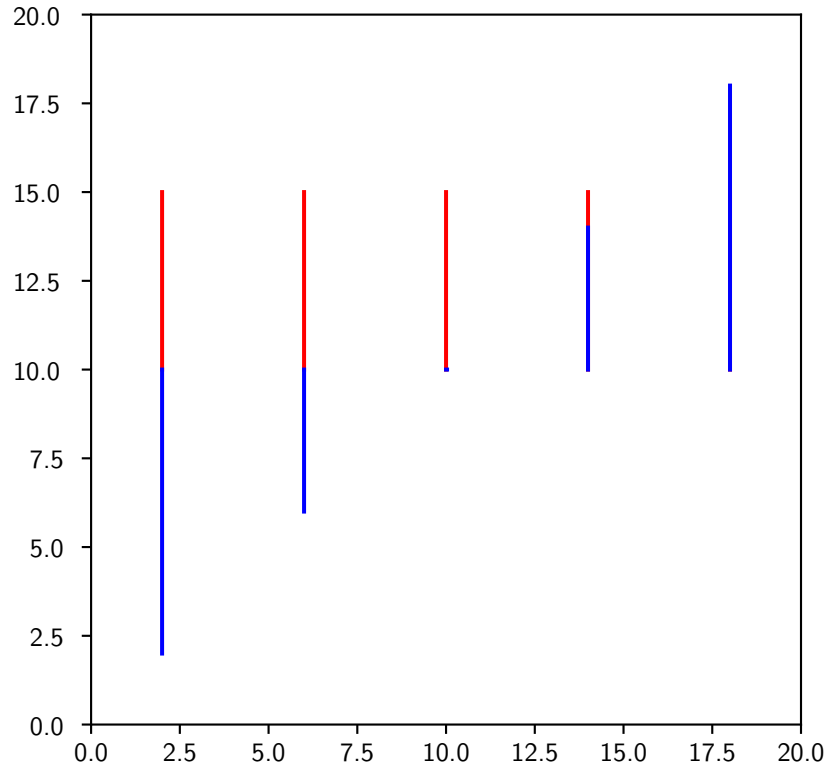
2 Question 10

- (a) We consider the following examples of exchange networks for Uber-like ridesharing apps, show their computed stable outcome, and argue the legitimacy of the computed driver profits and rider prices. We define the driver profits as their allocation in the stable outcome, and the rider prices as the difference between their valuation of reaching their destination and their allocation in the stable outcome.

Example 1 Depicted by the following grid representation, where red circles represent riders and blue circles represent drivers (note that the center of the grid has a rider and a driver yet only the driver is shown), and where red lines from the rider location (red circle) to the destination (green circle) represent the path the rider would like to take:



The matching of the stable outcome illustrated by connecting a driver to the rider by a blue line, and by connecting the rider's location to the destination by a red line, for each edge between a rider and a driver:



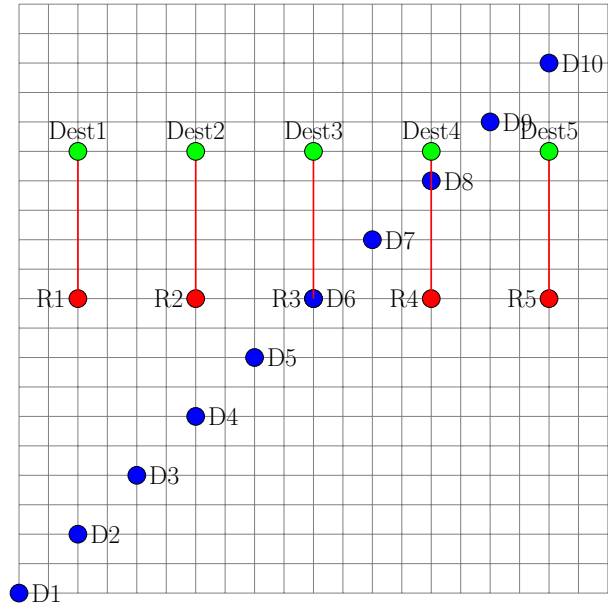
And the driver profits and rider prices for this stable outcome are as follows:

- Driver Profits: $[0, 0, 0, 0, 0]$
- Rider Prices: $[13, 9, 5, 9, 13]$

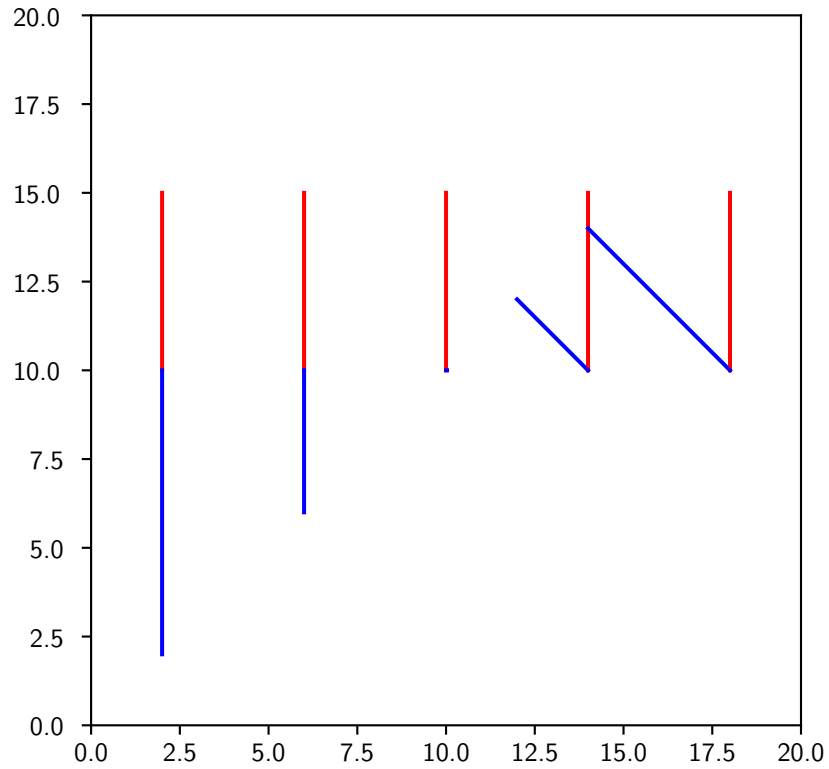
The profits are sensible, as the stable outcome is calculated by the the market equilibrium algorithm which prioritizes lower prices for the buyers—which we modeled to be the riders when encoding the exchange network as a matching market context.

FIXME: FILL IN MORE

Example 2 Depicted by the following grid representation is the second example, using similar conventions as the first example:



The matching of the stable outcome illustrated by connecting a driver to the rider by a blue line, and by connecting the rider's location to the destination by a red line, for each edge between a rider and a driver:



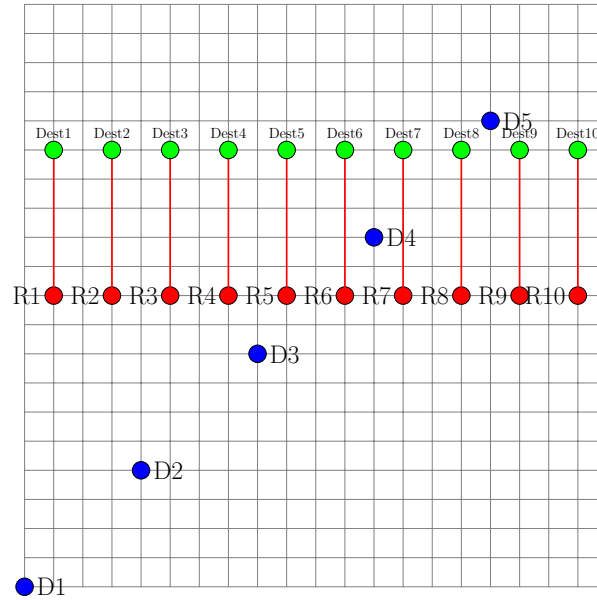
And the driver profits and rider prices for this stable outcome are as follows:

- Driver Profits: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
- Rider Prices: [13, 9, 5, 9, 13]

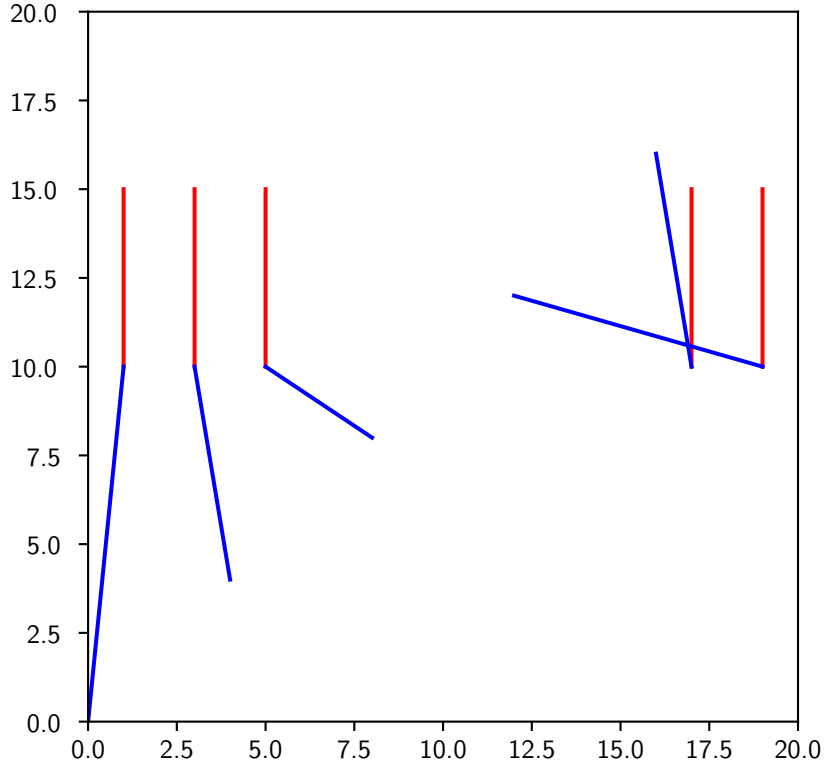
The profits are sensible, as the stable outcome is calculated by the the market equilibrium algorithm which prioritizes lower prices for the buyers—which we modeled to be the riders when encoding the exchange network as a matching market context.

FIXME: FILL IN MORE

Example 3 Depicted by the following grid representation is the third example, using similar conventions as the first example:



The matching of the stable outcome illustrated by connecting a driver to the rider by a blue line, and by connecting the rider's location to the destination by a red line, for each edge between a rider and a driver:



And the driver profits and rider prices for this stable outcome are as follows:

- Driver Profits: [19, 25, 27, 25, 25]
- Rider Prices: [35, 37, 37, 35, 30, 25, 30, 35, 37, 39]

The profits are sensible, as the stable outcome is calculated by the the market equilibrium algorithm which prioritizes lower prices for the buyers—which we modeled to be the riders when encoding the exchange network as a matching market context.

FIXME: FILL IN MORE

- (b) We implement a procedure that on a 100×100 grid, generates an exchange network context of n riders and m drivers with random locations and destinations, where each rider has value 100, and computes the stable outcome for this context. The following is the analysis of the results when running over different configurations of (n, m) for many times.

Configuration 1: $n = m = 10$

Configuration 2: $5 = n < m = 20$

3 Question 11

We can increase an edge's value if the destination is close to a high-value location.

This can be done in many ways, for example by lowering the value with the distance from the closest high-value location.

Such calculation can also incorporate the value of the high-value location:

$$\text{value decrease component by preference} = \min_{HVL \in \text{High-Value Locations}} \left(\frac{\|\text{rider location} - HVL\|_1}{\text{preference value of } HVL} \right)$$

4 Bonus Question 3

- (a) We simulated this as n more drivers simulating public transportation, whose cost of driving is the price described.

This ensures all players have the ability to take public transportation instead of uber.

Since we still have a bipartite graph, we know that a stable matching must exist.

- (b) Higher values for a means public transportation becomes less attractive for everyone.

Higher values for b means public transportation is more reliable for shorter distances.

$b > 1$ means the drive itself costs more with public transportation than with uber (so with a short distance between an uber and a rider it could be preferable to take an uber, depending on the base fare a).

References